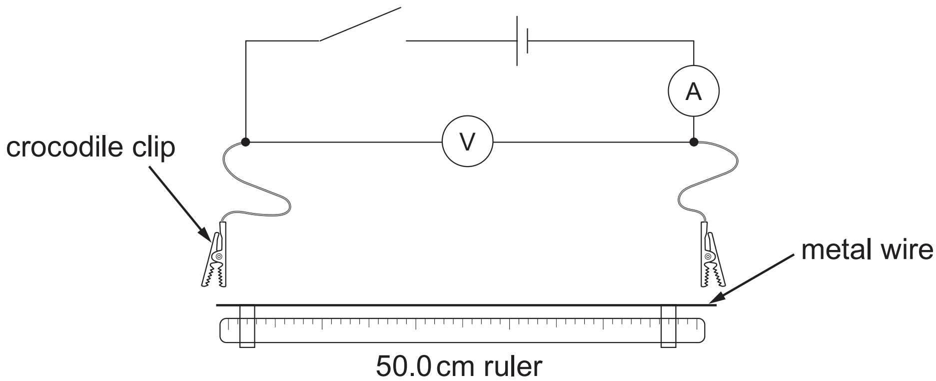
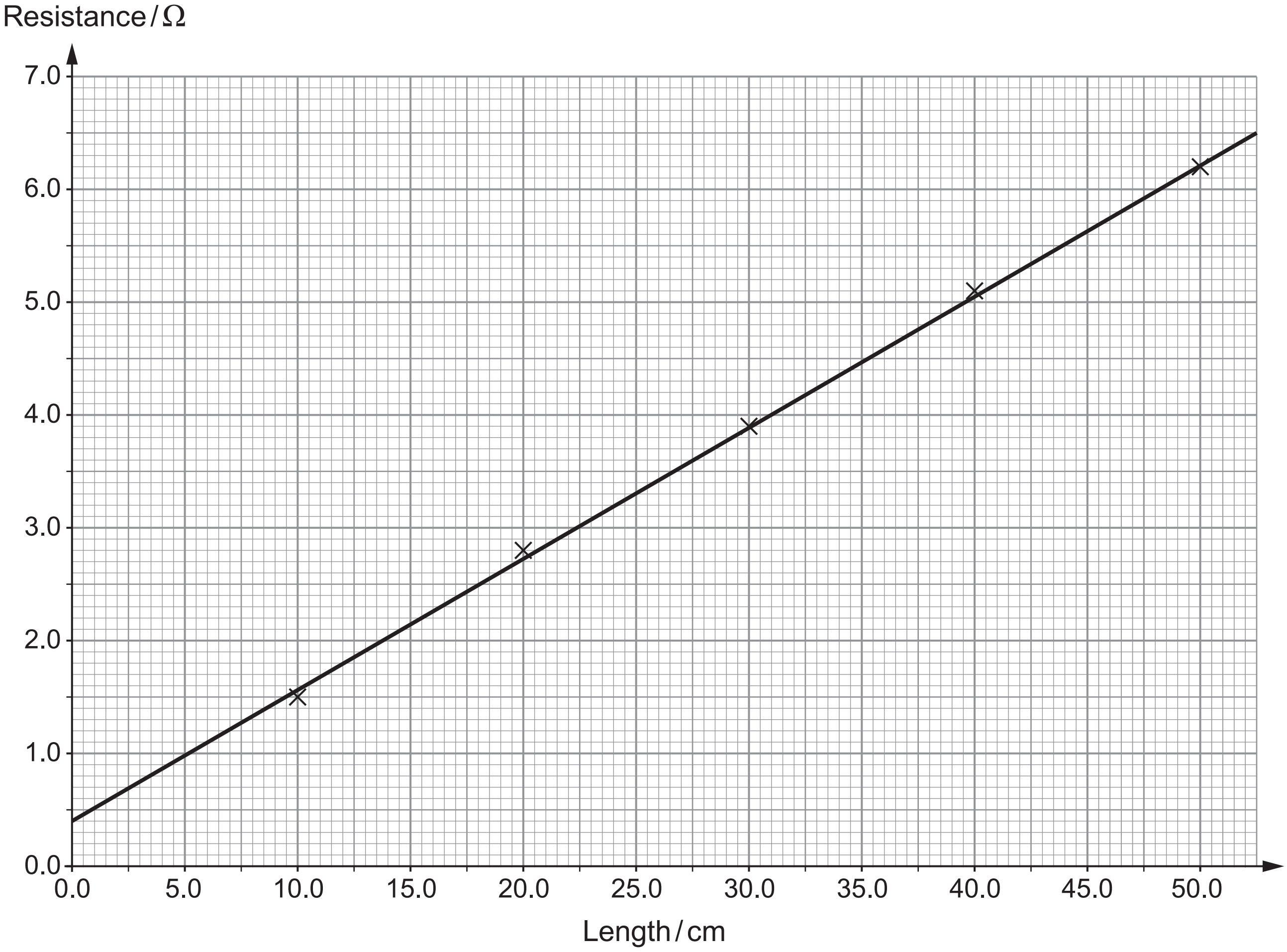
Answer **all** questions.

1. (a) Emily sets up the following circuit in order to investigate how the resistance of a metal wire varies with length.



Emily connects one crocodile clip on the wire at the 0.0 cm mark while the other clip is moved along at suitable intervals to cover the whole range of the wire. She obtains results as shown in the graph below.



Examiner only

(i) Use the graph to obtain a value for the resistance at 0.0 cm and suggest the cause of this resistance. [2]

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(ii) Discuss to what extent Emily's results confirm that the variation of resistance of the wire with length is consistent with the equation: [3]

$$R = \frac{\rho l}{A}$$

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(iii) The wire used in the experiment has a mean **diameter** of 0.23 mm. Name a measuring instrument that could have been used to take this reading **and** state its likely resolution. [2]

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(iv) Use the graph and the mean diameter to determine the resistivity of the metal of the wire. [4]

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(b) Emily used a 1.5 V cell in her experiment as she was concerned about the size of the electrical current in the wire. Suggest her reasons for using the 1.5 V cell. [2]

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